

6-Month MMANGO Interview Prep Plan (2-5 Years Experience)

MY DSA NOTES I USED TO CRACK MICROSOFT:

<https://topmate.io/itshimanshu007/1715501>

Month 1: Microsoft + Coding Fundamentals

Week	Focus Area	Topics to Study	Practice/Action Items	Resources
Week 1	Data Structures	Arrays, Strings, HashMaps, Sets	- Solve 15 easy problems on LeetCode - Focus on string manipulation, two pointers - Practice time/space complexity analysis	LeetCode, NeetCode 150
Week 2	Data Structures	Linked Lists, Stacks, Queues, Trees	- Solve 15 medium problems - Implement data structures from scratch - Practice recursive tree traversals	LeetCode, Cracking the Coding Interview
Week 3	Microsoft Products	Azure basics, M365, .NET ecosystem	- Create free Azure account, build simple app - Learn C# basics if not familiar - Study Azure services (VMs, Functions, Storage)	Microsoft Learn, Azure Documentation
Week 4	System Design Intro	Scalability basics, Load balancing, Caching	- Read "Designing Data-Intensive Applications" (Ch 1-3) - Watch System Design Primer videos - Design: URL shortener, Pastebin	System Design Primer (GitHub)

Weekly Goal: 20-25 coding problems + 1 system design problem + 1 Microsoft product exploration

Month 2: Meta + Advanced DSA

Week	Focus Area	Topics to Study	Practice/Action Items	Resources
Week 1	Algorithms	Binary Search, Sorting, Sliding Window	- Solve 20 medium problems - Master binary search variations - Practice window-based problems	LeetCode patterns, AlgoExpert
Week 2	Algorithms	DFS, BFS, Backtracking	- Solve 20 medium/hard graph problems - Practice matrix traversal - Solve permutations/combinations	LeetCode, Blind 75
Week 3	Meta Products	React, GraphQL, News Feed architecture	- Build small React app - Study Meta's tech stack (React, PyTorch, Hack) - Learn about social graph concepts	React docs, Meta Engineering Blog
Week 4	System Design	Database design, SQL vs NoSQL, CAP theorem	- Design: Instagram, Facebook News Feed - Practice drawing architecture diagrams - Study data modeling	Grokking System Design

Weekly Goal: 25-30 coding problems + 1 system design problem + Meta tech exploration

Month 3: Apple + Object-Oriented Design

Week	Focus Area	Topics to Study	Practice/Action Items	Resources
Week 1	Advanced DSA	Dynamic Programming (1D, 2D)	- Solve 25 DP problems (easy to hard) - Master memoization and tabulation - Study classic DP patterns	LeetCode DP patterns, NeetCode
Week 2	Advanced DSA	Graphs (Dijkstra, Union Find, Topological Sort)	- Solve 20 graph problems - Implement graph algorithms from scratch - Practice shortest path problems	LeetCode, Graph Theory resources
Week 3	Apple Products	Swift/Objective-C basics, iOS architecture, macOS	- Learn Swift fundamentals - Study Apple design principles (HIG) - Understand Apple Silicon, ecosystem integration	Swift documentation, Apple Developer
Week 4	OOD + Low Level Design	Design patterns, SOLID principles, UML	- Design: Parking Lot, Library System, Chess - Practice explaining design choices - Study creational/structural patterns	Head First Design Patterns

Weekly Goal: 30 coding problems + 1 LLD problem + 1 HLD problem + Apple tech study

Month 4: NVIDIA + Google (Core CS)

Week	Focus Area	Topics to Study	Practice/Action Items	Resources
Week 1	Advanced Algorithms	Greedy, Heaps, Trie, Bit Manipulation	- Solve 25 problems across topics - Focus on optimization problems - Practice bitwise operations	LeetCode, Competitive Programming
Week 2	NVIDIA Focus	CUDA basics, GPU architecture, Parallel computing	- Learn basic CUDA programming - Study GPU vs CPU architecture - Understand deep learning frameworks	NVIDIA Developer, CUDA documentation
Week 3	Google DSA	Hard problems, Google interview patterns	- Solve 30 medium-hard problems - Focus on Google-tagged problems - Practice optimization techniques	LeetCode (Google tagged)
Week 4	System Design	Distributed systems, Microservices, Message Queues	- Design: Google Search, YouTube, Gmail - Study MapReduce, GFS, Bigtable papers - Learn Kafka, RabbitMQ basics	Google SRE Book, Research Papers

Weekly Goal: 35-40 coding problems + 2 system design problems + domain study

Month 5: OpenAI + Mock Interviews

Week	Focus Area	Topics to Study	Practice/Action Items	Resources
Week 1	ML/AI Basics	Neural Networks, Transformers, LLMs overview	- Study ML fundamentals (supervised/unsupervised) - Understand transformer architecture basics - Learn about GPT, embeddings, fine-tuning	Fast.ai, 3Blue1Brown, Papers
Week 2	OpenAI Focus	API usage, Prompt engineering, Model evaluation	- Build project using OpenAI API - Practice prompt engineering - Study RLHF, alignment concepts	OpenAI Documentation, Cookbook
Week 3	Mock Interviews	Coding rounds simulation	- 4 mock coding interviews (Pramp, Interviewing.io) - Record yourself, review performance - Work on communication and clarity	Pramp, Interviewing.io
Week 4	Mock Interviews	System design rounds simulation	- 3 mock system design interviews - Practice whiteboard/virtual whiteboarding - Get feedback from peers/mentors	Exponent, SystemsExpert

Weekly Goal: 20 coding problems + 2 mocks + ML concepts + project building

Month 6: Behavioral + Final Prep

Week	Focus Area	Topics to Study	Practice/Action Items	Resources
Week 1	Behavioral Prep	STAR method, Leadership principles	- Write 15-20 STAR stories covering: * Conflict, failure, leadership, impact * Cross-team collaboration, trade-offs - Map stories to company values	Company career pages
Week 2	Company-Specific Prep	Culture, values, recent news	- Study each company's leadership principles: * Microsoft: Growth mindset, customer obsession * Meta: Move fast, be bold * Apple: Think different, attention to detail * Google: Googleyness, innovation - Research recent products/launches	Glassdoor, Blind, company blogs
Week 3	Final Coding Sprint	Weak areas, hard problems	- Identify weakest topics, solve 40 problems - Revisit hardest problems from past months - Time yourself strictly (45 min/problem)	LeetCode, previous mistakes log
Week 4	Final Mocks + Review	Full interview simulations	- 2 full mock interview loops (4-5 rounds each) - Practice system design on whiteboard	Mock platforms, study notes

			- Review all notes and patterns - Rest 2 days before actual interviews	
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Weekly Goal: Behavioral mastery + 25 problems + company research + final polish

Daily Schedule (Throughout 6 Months)

Time Slot	Activity	Duration
Morning	Coding Problems (DSA)	1.5-2 hours
Afternoon	System Design / Domain Study	1 hour
Evening	Company-specific learning / Product exploration	45 min
Night	Review mistakes, update notes	30 min

Total: 3.5-4 hours/day

Key Metrics to Track

Metric	Target
Total LeetCode Problems	400+ (150 Easy, 200 Medium, 50 Hard)
System Design Problems	30+
Low Level Design Problems	15+
Mock Interviews	10+
STAR Stories Prepared	20+
Projects Built	3-4 (one per major company focus)

Company-Specific Interview Focus

Company	Coding Difficulty	System Design Weight	Behavioral Focus	Special Areas
Microsoft	Medium	Medium	Growth mindset, collaboration	Azure, C#/.NET, cloud

Meta	Medium-Hard	High	Impact, moving fast	React, distributed systems
Apple	Medium	Low-Medium	Attention to detail, polish	Swift, OOD, hardware-software
NVIDIA	Medium-Hard	Medium	Innovation, technical depth	CUDA, parallel computing, graphics
Google	Hard	High	Googleyness, collaboration	Scalability, algorithms
OpenAI	Medium-Hard	Medium	Alignment, research mindset	ML/AI, transformers, ethics